



HK IMO Training 2021-2022
Problem Set 2
For 2 October 2021



Problem 5. There are $n > 2$ nonzero integers such that each of them is divisible by the sum of the other $n - 1$ numbers. Prove that the sum of all these integers is 0.

Problem 6. Let S be a set of $2k$ elements where k is a positive integer. Let $\mathcal{F}$ be a family of subsets of S such that every set in $\mathcal{F}$ has exactly k elements. Suppose every subset of S having exactly $k - 1$ elements is a subset of exactly one element in $\mathcal{F}$. Prove that $k + 1$ is a prime number.

Problem 7. Let $a_1, a_2, \dots, a_n$ be positive real numbers satisfying $a_1^2 + 2a_2^3 + \dots + na_n^{n+1} < 1$. Prove that

$$2a_1 + 3a_2^2 + \dots + (n + 1)a_n^n < 3.$$

Problem 8. Let $\triangle ABC$ be a scalene acute triangle. Let D be a point on the side BC . Points E and F are chosen on the sides AB and AC respectively such that $\angle DEB = \angle DFC$. Define the following.

- P is the intersection point of the extensions of FD and AB
- Q is the intersection point of the extensions of ED and AC
- Γ_1 is the circumcircle of $\triangle DEP$
- Γ_2 is the circumcircle of $\triangle DFQ$
- Ω_1 is the circle internally tangent to Γ_1 at D and tangent to the line AB at S
- Ω_2 is the circle internally tangent to Γ_2 at D and tangent to the line AC at T
- X is the intersection point of Γ_1 and Γ_2 different from D
- Y is the intersection point of Ω_1 and Ω_2 different from D
- K is the intersection point of the circumcircles of $\triangle AEF$ and $\triangle AST$ different from A
- L is the intersection point of the circumcircle of $\triangle AEF$ and the line AY different from A

(a) Prove that D, X, Y are collinear.

(b) Prove that the tangent at D to the circumcircle of $\triangle DYK$ meets EF at a point on the circumcircle of $\triangle DKL$.